

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2088

Opequon 138 kV
June 02, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2088. The Interconnection Customer proposes a 169.5 MW solar photovoltaic facility interconnecting at the Opequon 138 kV substation in the West Virginia area. The request was received on May 20, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 703 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,947,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Opequon-Lcec No. 5 138kV line
Contingency: Clark County-Kearneyville 345kV
Loading: 130.8% (Rating: 515 MVA)
DFAX: 42.6%
Pre-project loading: 116.8%
2. Facility: Jack McDonough-Low Gap 138kV Line
Contingency: Artesia Country Club-Clark County Substation 345kV line
Loading: 117.1% (Rating: 581 MVA)
DFAX: 6.8%
Pre-project loading: 115.1%
3. Facility: Arvin-Shuffleton 138kV
Contingency: Kearneyville-Clark County 345kV L
Loading: 116.3% (Rating: 277 MVA)
DFAX: 12.9%
Pre-project loading: 108.4%
4. Facility: Smallwood-Killingsworth Substation 138kV Line
Contingency: Kearneyville-Clark County 345kV
Loading: 114.9% (Rating: 245 MVA)
DFAX: 9.1%
Pre-project loading: 108.5%
5. Facility: Columbus City 345/1kV Transformer
Contingency: Kearnyville-International Paper Valliant Ok 345kV line
Loading: 111.9% (Rating: 586 MVA)
DFAX: 30.8%
Pre-project loading: 103.0%

6. Facility: Columbus City-Smallwood 138kV line
Contingency: Clark County-Kearneyville 345kV
Loading: 111.0% (Rating: 267 MVA)
DFAX: 9.1%
Pre-project loading: 105.2%
7. Facility: Air Force Base 345/138kV Autotransformer
Contingency: Tiger Bay Substation-Air Fore Base 345kV
Loading: 110.0% (Rating: 704 MVA)
DFAX: 8.3%
Pre-project loading: 108.0%
8. Facility: Southwest Research Institute-Douglas City 138kV line
Contingency: Clark County-Kearneyville 345kV
Loading: 109.2% (Rating: 306 MVA)
DFAX: 9.3%
Pre-project loading: 104.1%
9. Facility: Columbus City 138-1 kV Transformer Bank
Contingency: Kearneyville-International Paper Valliant Ok Substation 345kV
Loading: 106.8% (Rating: 629 MVA)
DFAX: 30.8%
Pre-project loading: 98.5%
10. Facility: Frederick-Altavista Power Station 138kV
Contingency: Jack Mcdonough-Low Gap 138kV
Loading: 105.3% (Rating: 433 MVA)
DFAX: 6.2%
Pre-project loading: 102.9%
11. Facility: Columbus City-North Mesa 138kV line
Contingency: Kearneyville-International Paper Valliant Ok 345kV
Loading: 102.6% (Rating: 452 MVA)
DFAX: 39.2%
Pre-project loading: 87.9%
12. Facility: Clark County-Kearneyville 345kV
Contingency: Sleepy Hollow-Grimmway-Malaga 138kV
Loading: 101.7% (Rating: 2,021 MVA)
DFAX: 25.7%
Pre-project loading: 99.6%
13. Facility: Arvin-Southwest Research Institute 138kV
Contingency: Clark County-Kearneyville 345kV line
Loading: 101.5% (Rating: 292 MVA)
DFAX: 9.3%
Pre-project loading: 96.1%
14. Facility: Sleepy Hollow-Grimmway-Malaga 138kV
Contingency: Clark County-Kearneyville 345kV
Loading: 101.2% (Rating: 623 MVA)
DFAX: 15.4%
Pre-project loading: 97.0%
15. Facility: Claytonia-Shuffleton 138kV line
Contingency: Kearneyville-Clark County 345kV line
Loading: 101.1% (Rating: 283 MVA)
DFAX: 12.9%
Pre-project loading: 93.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Opequon-Lcec No. 5 138kV line	Rebuild line with higher-capacity conductor	\$4,655,000	22% (\$1,012,000)
Jack McDonough-Low Gap 138kV Line	Rebuild line with higher-capacity conductor	\$1,623,000	10% (\$167,000)
Arvin-Shuffleton 138kV	Rebuild line with higher-capacity conductor	\$1,475,000	29% (\$422,000)
Smallwood-Killingsworth Substation 138kV Line	Rebuild line with higher-capacity conductor	\$4,256,000	12% (\$522,000)
Columbus City 345/1 kV TX	Replace transformer with higher-rated unit	\$12,833,000	24% (\$3,131,000)
Columbus City-Smallwood 138kV line	Rebuild line with higher-capacity conductor	\$7,992,000	12% (\$981,000)
Air Force Base 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$12,375,000	6% (\$779,000)
Douglas City-Southwest Research Institute 138kV	Reconductor line	\$1,525,000	59% (\$903,000)
Columbus City 138/1kV XFMR	Replace transformer with higher-rated unit	\$6,383,000	24% (\$1,557,000)
Frederick-Altavista Power Station 138kV	Reconductor line	\$5,487,000	100% (\$5,487,000)
Columbus City-North Mesa 138kV	Reconductor line	\$527,000	80% (\$424,000)
Clark County-Kearneyville Substation 345kV	Reconductor line	\$18,476,000	4% (\$750,000)
Arvin-Southwest Research Institute 138kV	Reconductor line	\$2,425,000	59% (\$1,436,000)
Sleepy Hollow-Grimmway-Malaga 138kV	Reconductor line	\$634,000	15% (\$94,000)
Claytonia-Shuffleton 138kV	Reconductor line	\$985,000	29% (\$282,000)
	Total allocated to project		\$17,947,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.